PhD Research Plan

Deep Learning for External Vehicle Aerodynamics – 2y Plan

Audi AG / TUM - Chair of Physics-Based Simulation
May 2026 - Jan 2029

Research direction

The target is not another model with a slightly better average error. I aim to build a reliable aerodynamic AI workflow: one that predicts well, solves the limits of previous methods, and can continuously learn from data of different fidelity.

1 → 2 → 3 → ?

1 Failure Analysis

Find where models fail

2 Robust Foundation model

Fix failures and design a reliable architecture

3 Pre-training, Fine-tuning and Active Learning

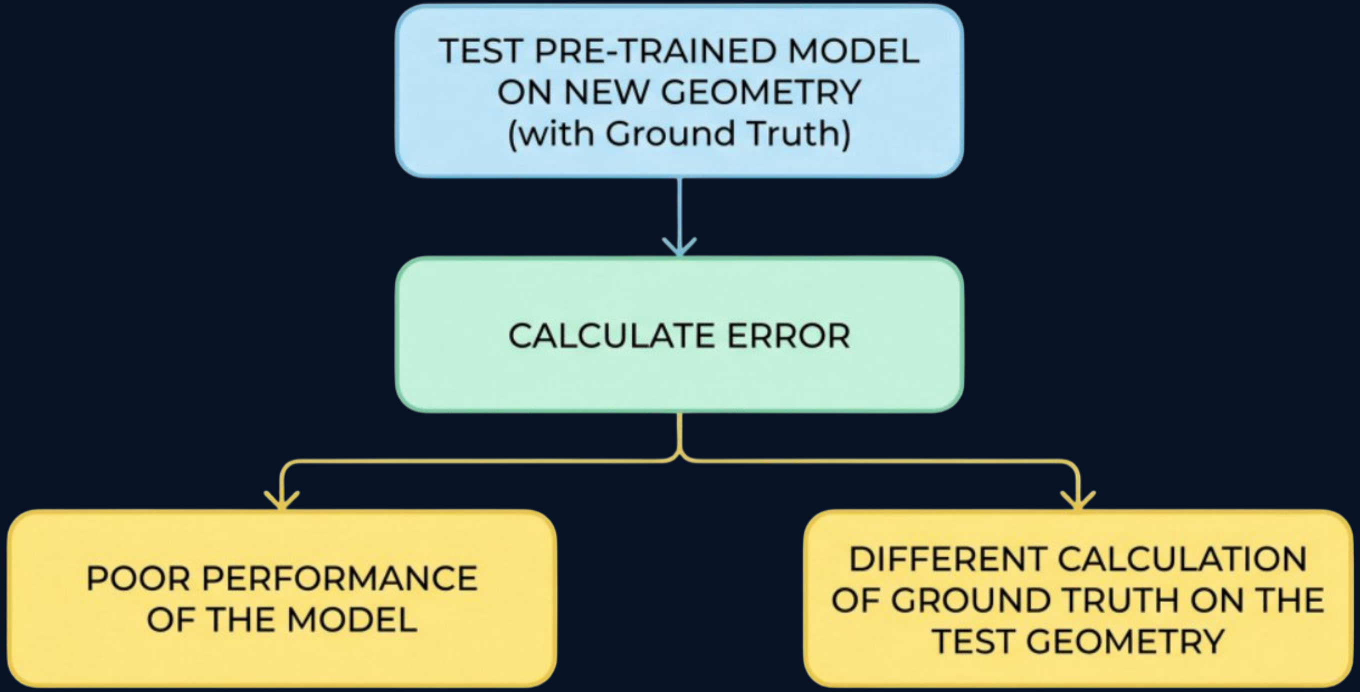
Learn reusable geometry-flow priors

Objective 1: Benchmarking and failure analysis

The first technical task is to study the performance of SOTA models under different scenarios. The purpose is not only to find the lowest error, but to understand where each model is unreliable.

EXPECTED OUTPUT

A failure taxonomy that defines which weaknesses must be solved as a priority.



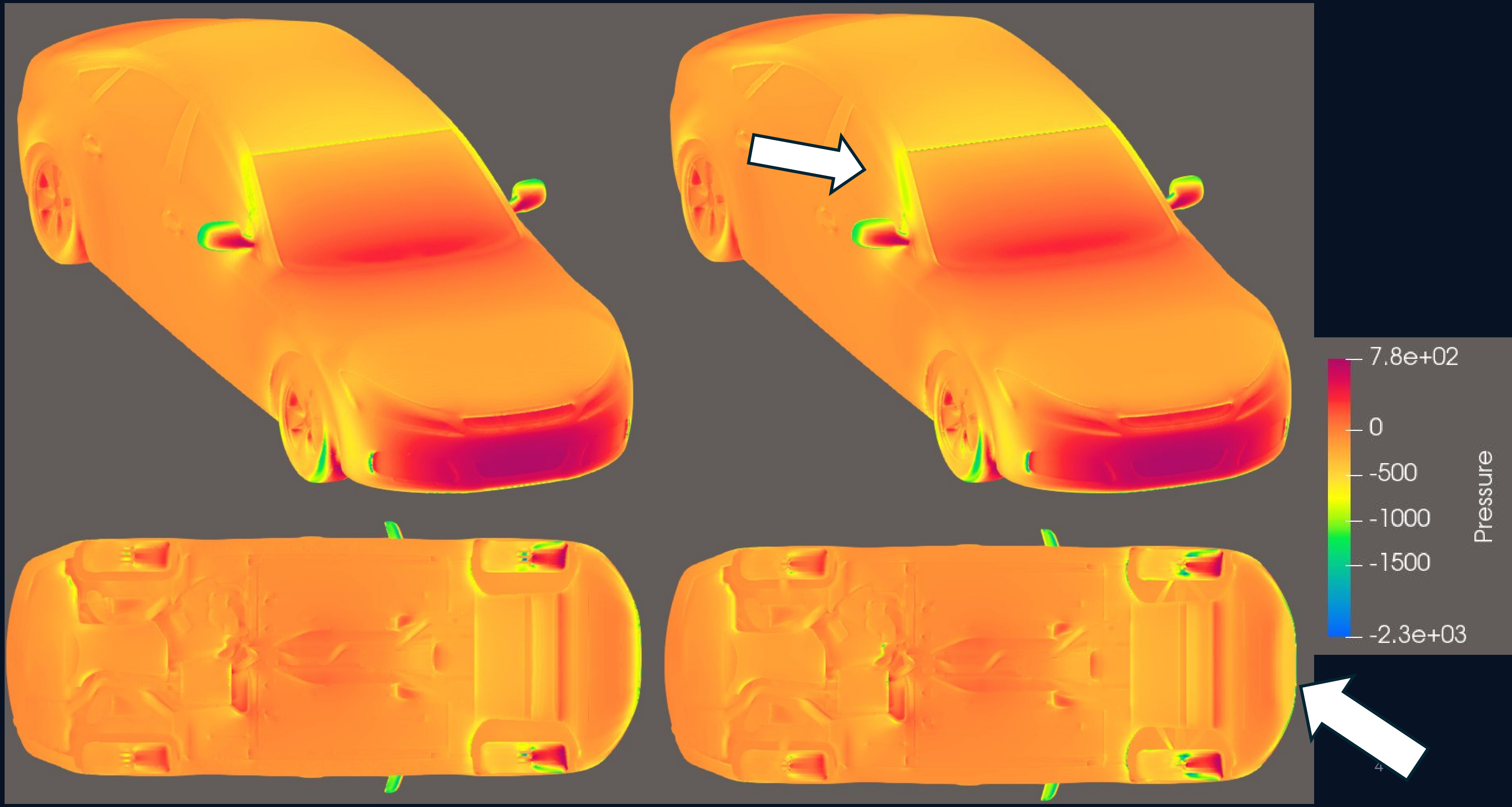
Work package 1.1: Difference between public datasets and Audi simulations

Surface Pressure using Audi Setup: left
Vs. Public dataset: right

Cd difference: ~1.5%

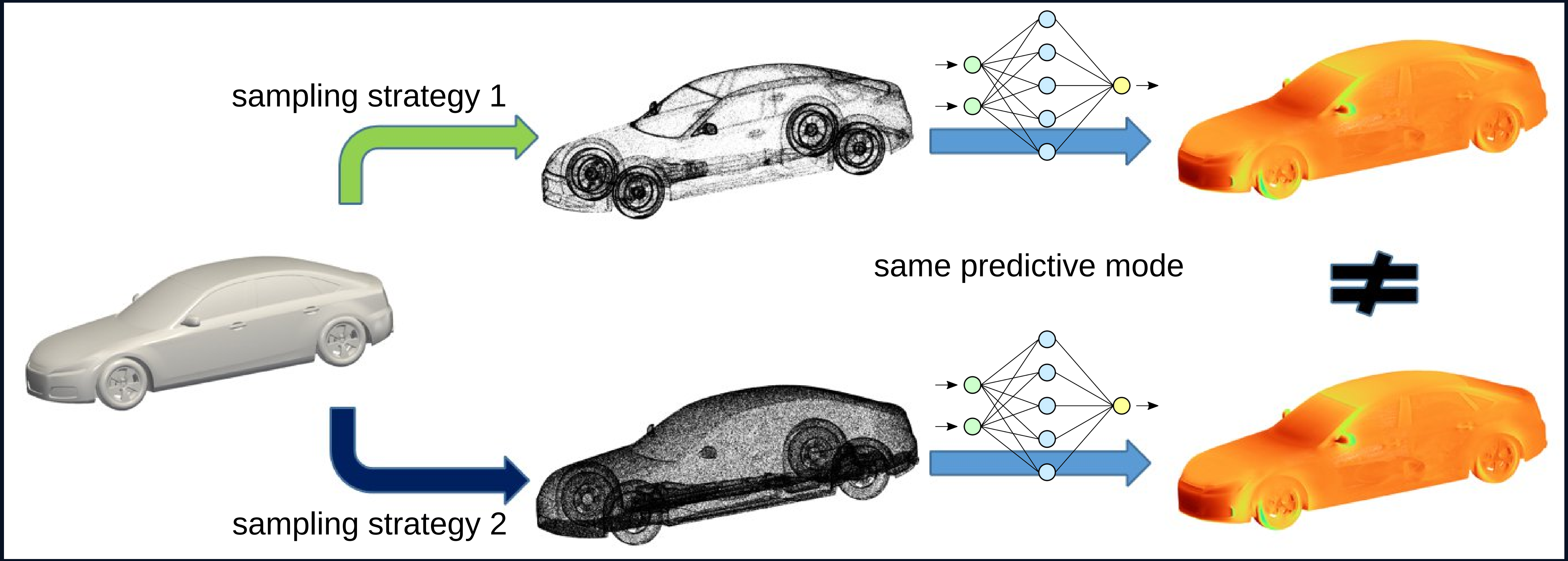
Audi CFD

Public Dataset

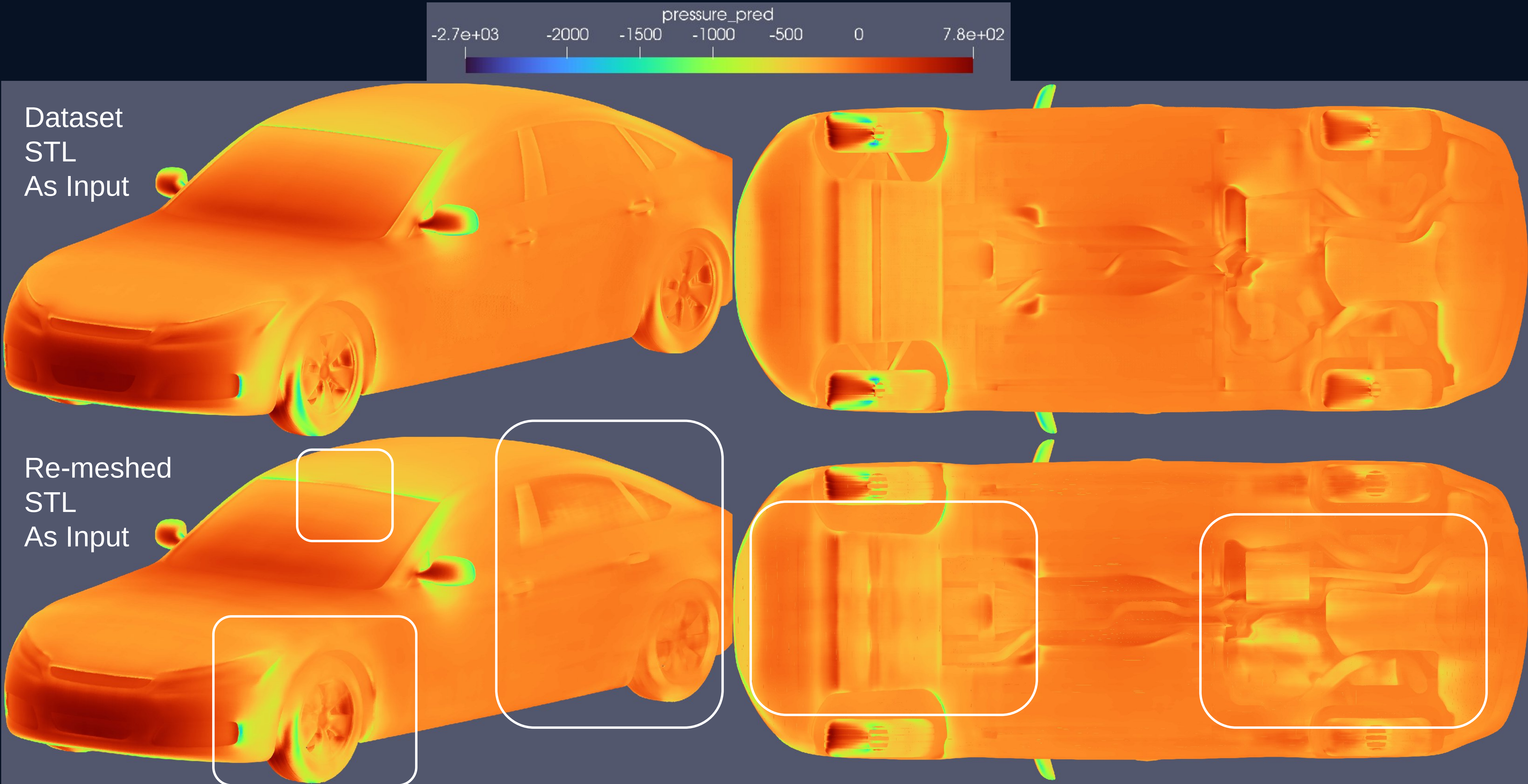


Work package 1.2: Sampling Sensitivity of Current Models

An AI model works well on its training data. But when you re-mesh the same geometries, the performance degrades sharply. This also happens by changing how you sample points for network input.



Predictions from a SOTA model on one of the STL files it was trained on

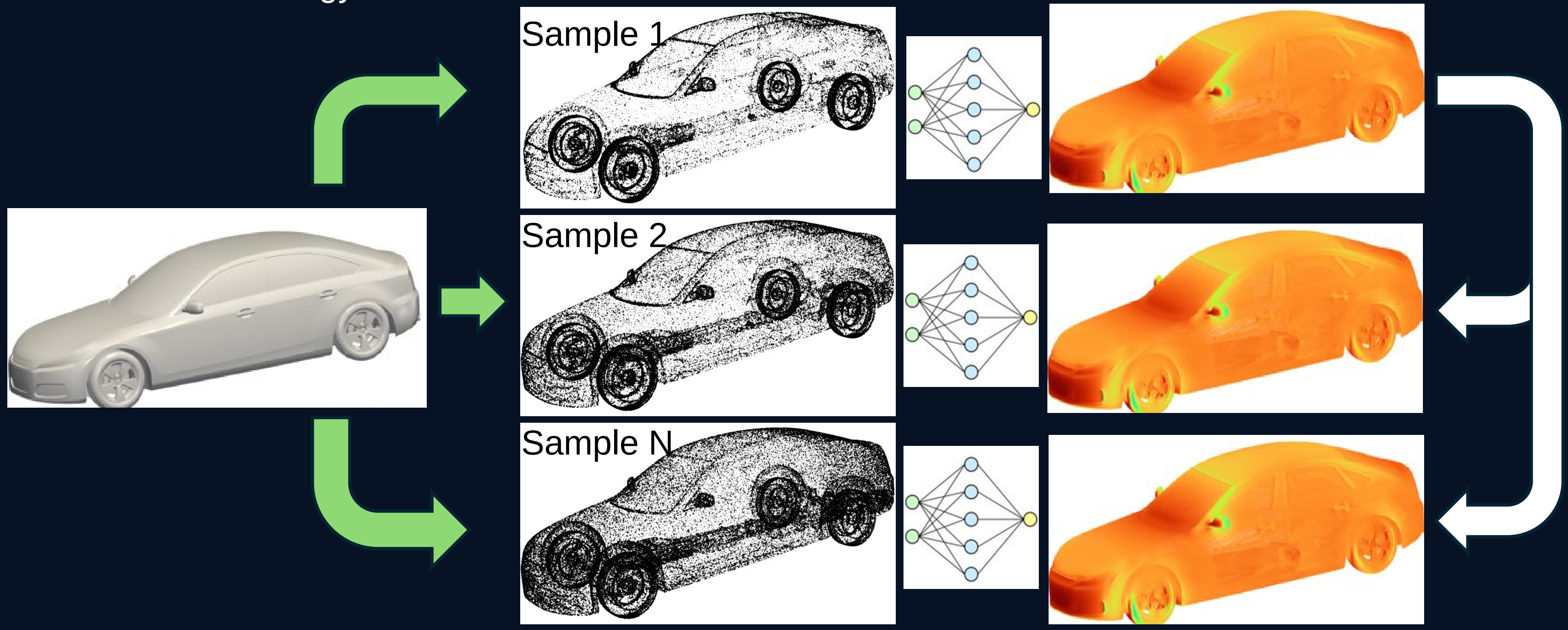


Work package 1.2: our solution

Different sampling or remeshing strategy

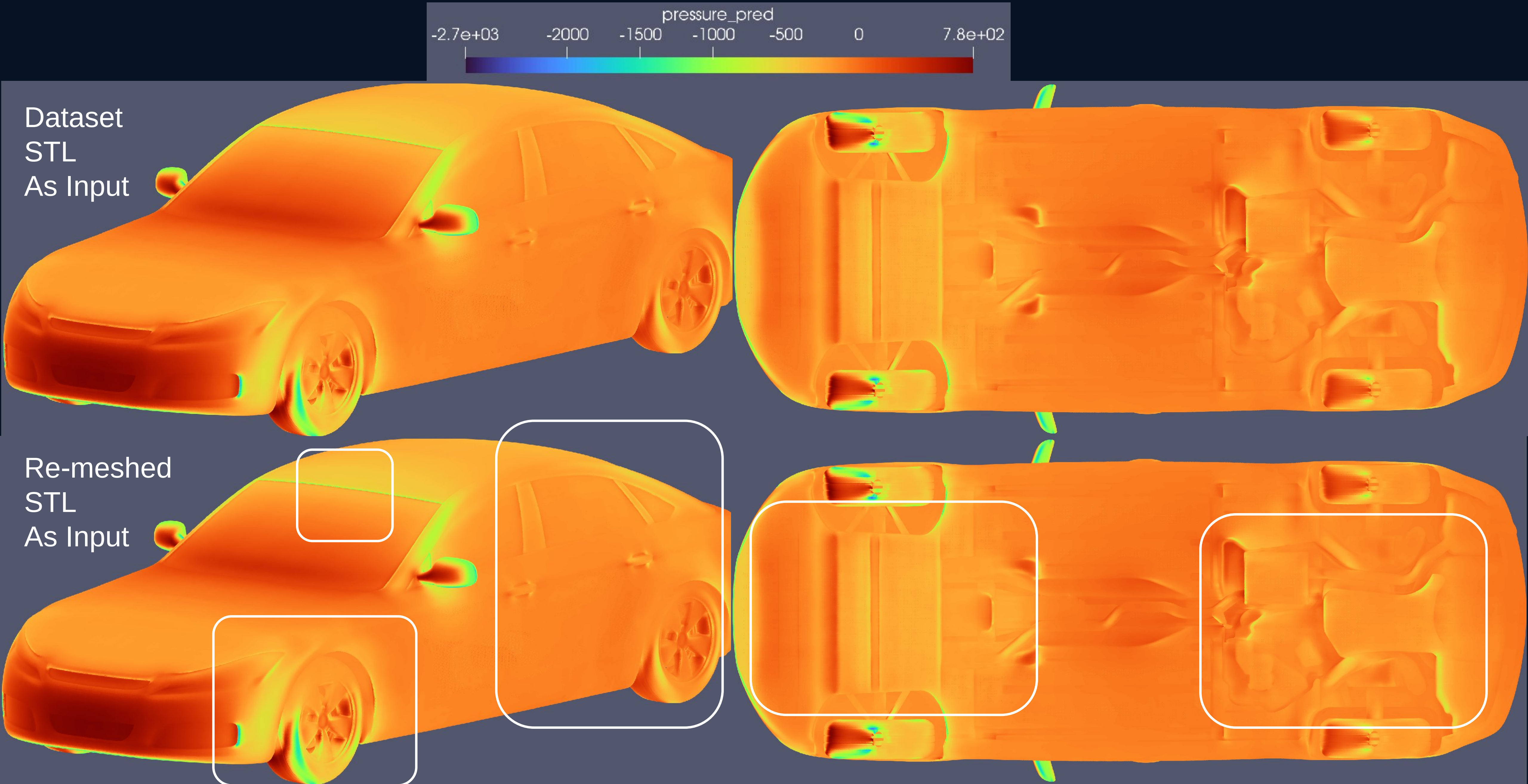
Making predictions

Comparing predictions in a loss function



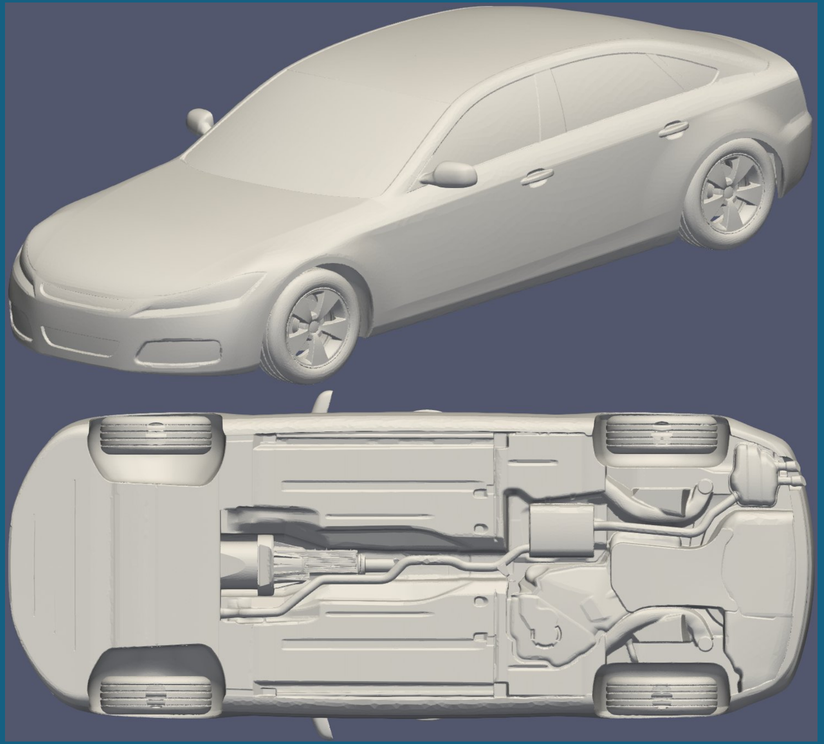
We penalize the model for making different predictions when the original geometry is the same but point sampling strategies are different

Work package 1.2: Results

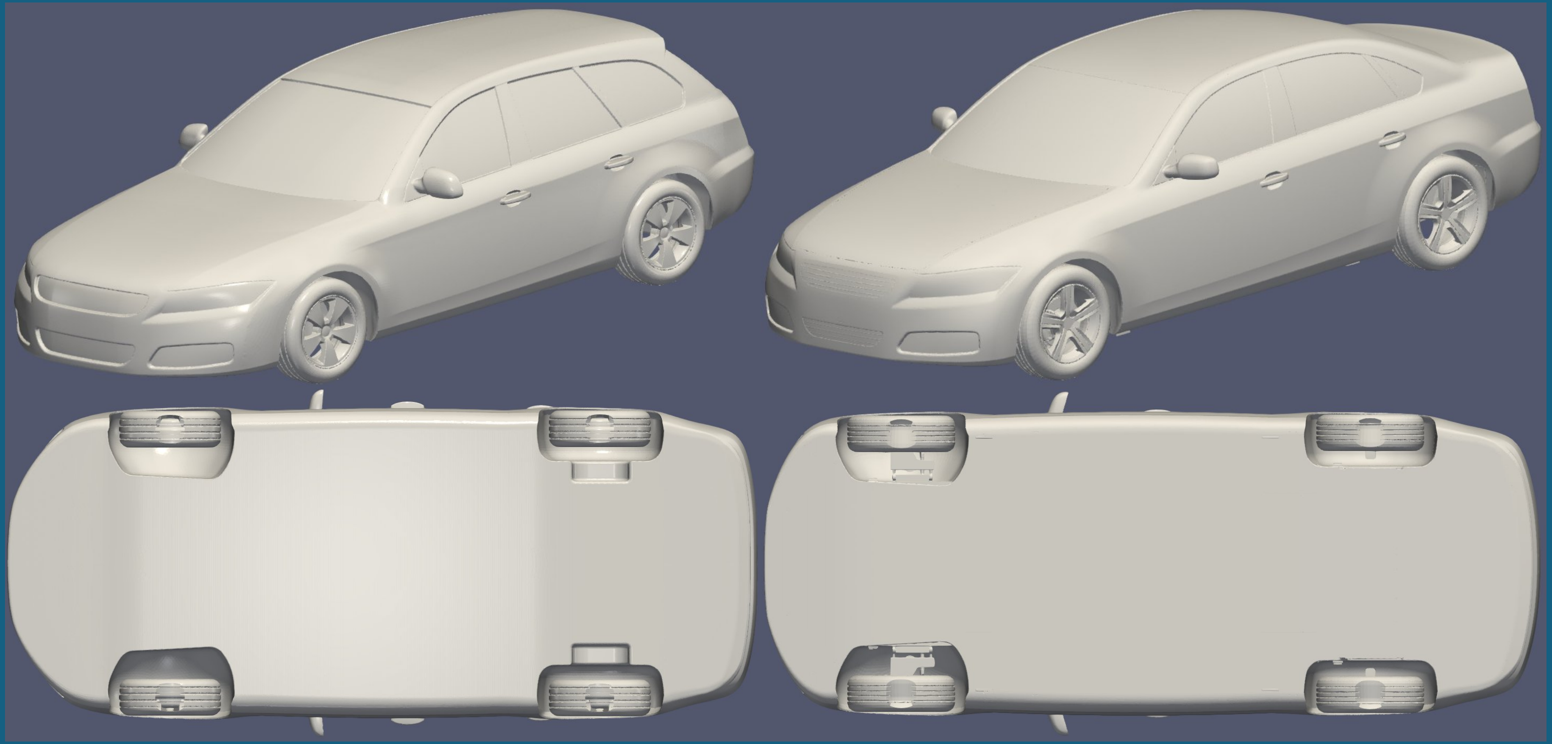


Work package 1.3: out of distribution input to pre-trained models

Most AI model works well on its training data. But when a new type of geometry that it is hasn't seen before is presented to it the results start to deteriorate and become meaningless outputs.

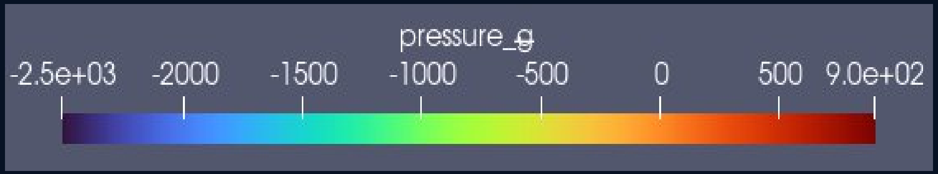


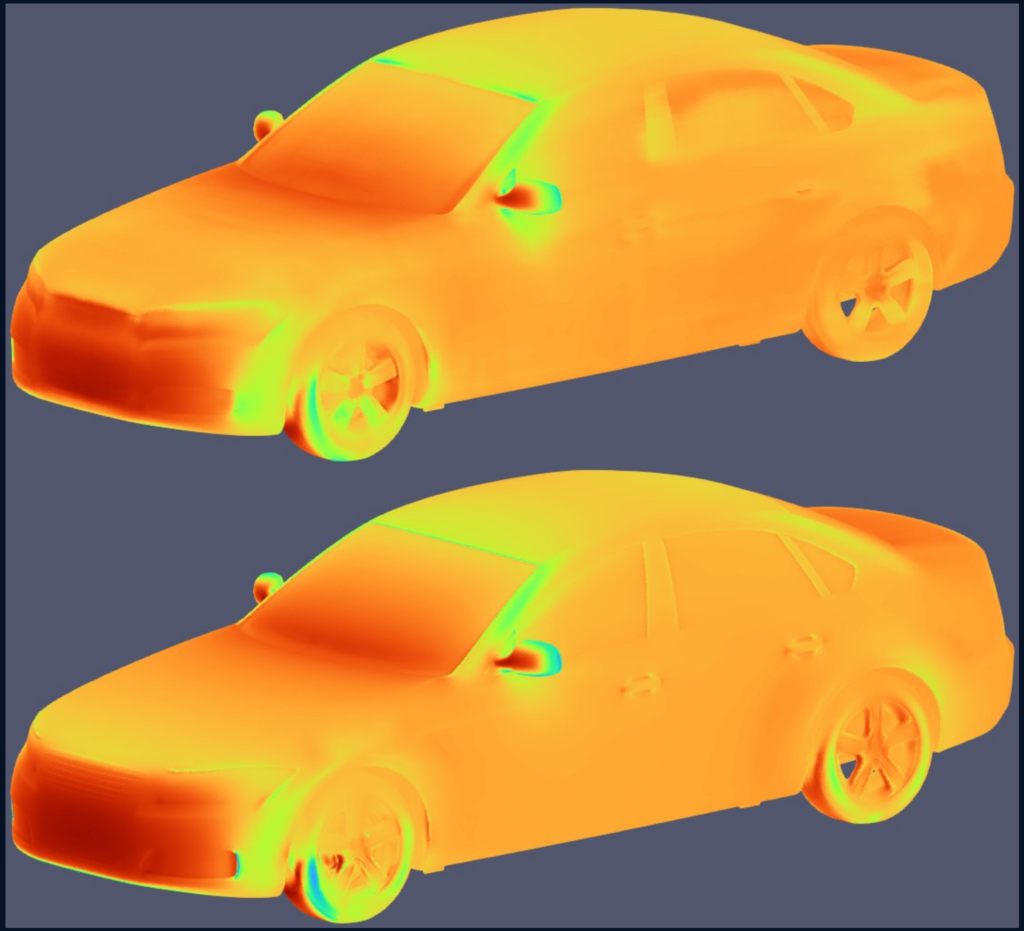
The Style of Training Geometries



The Style of Testing Geometries

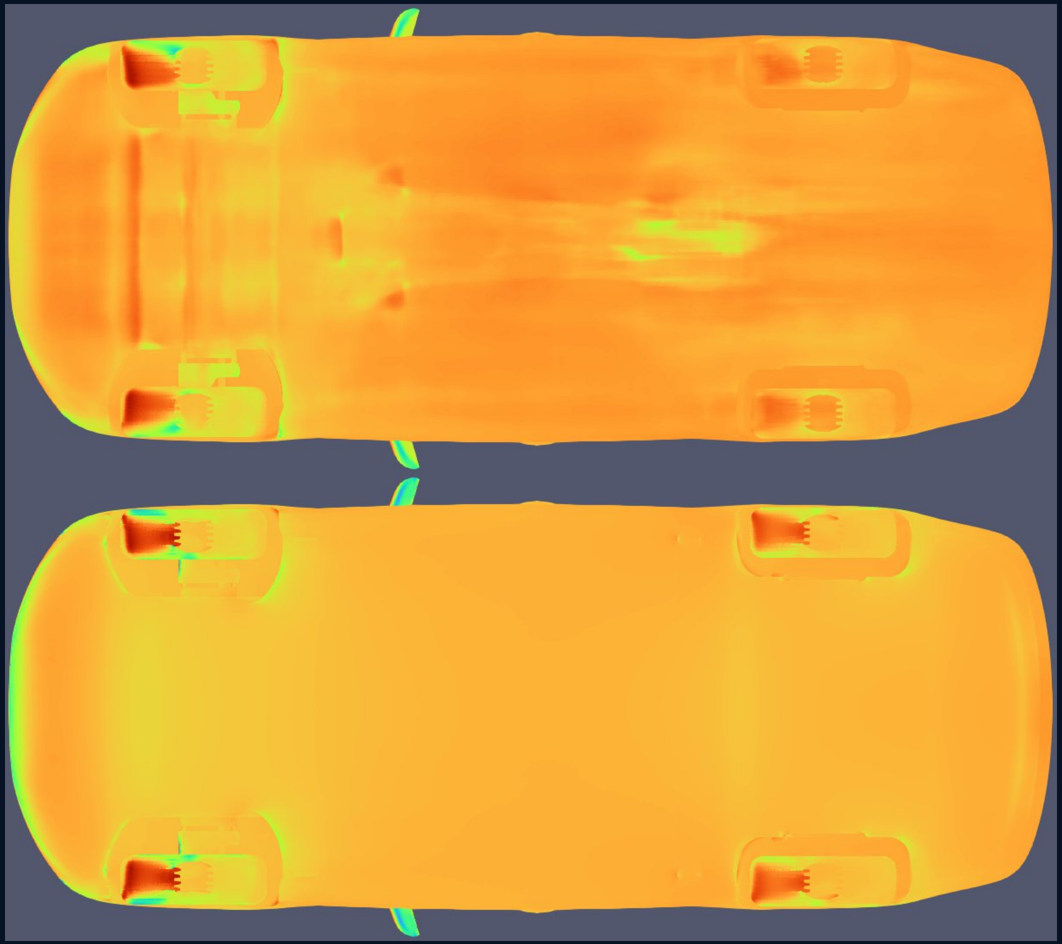
Work package 1.3: Evidence of the issue



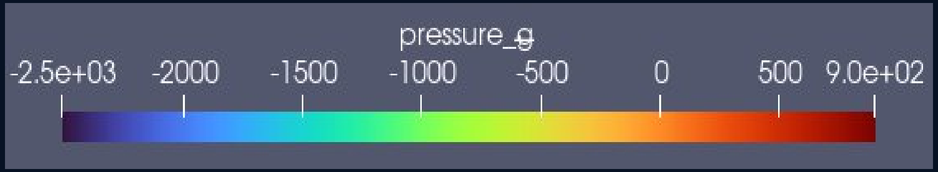


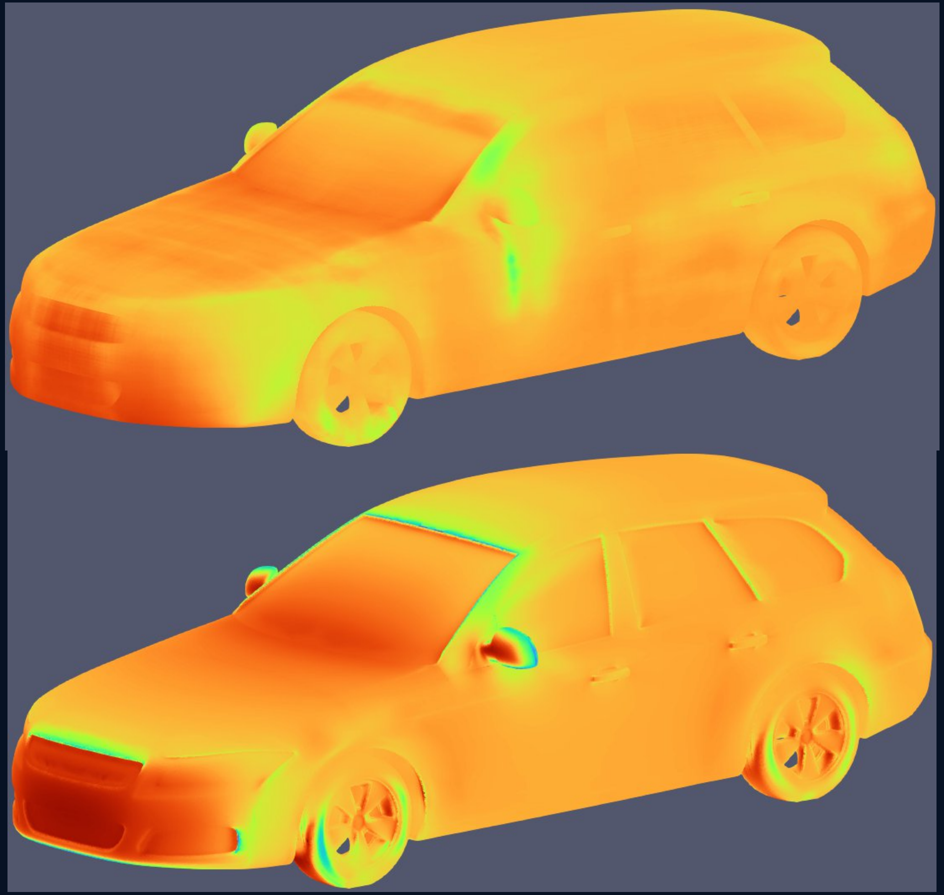
Prediction

CFD



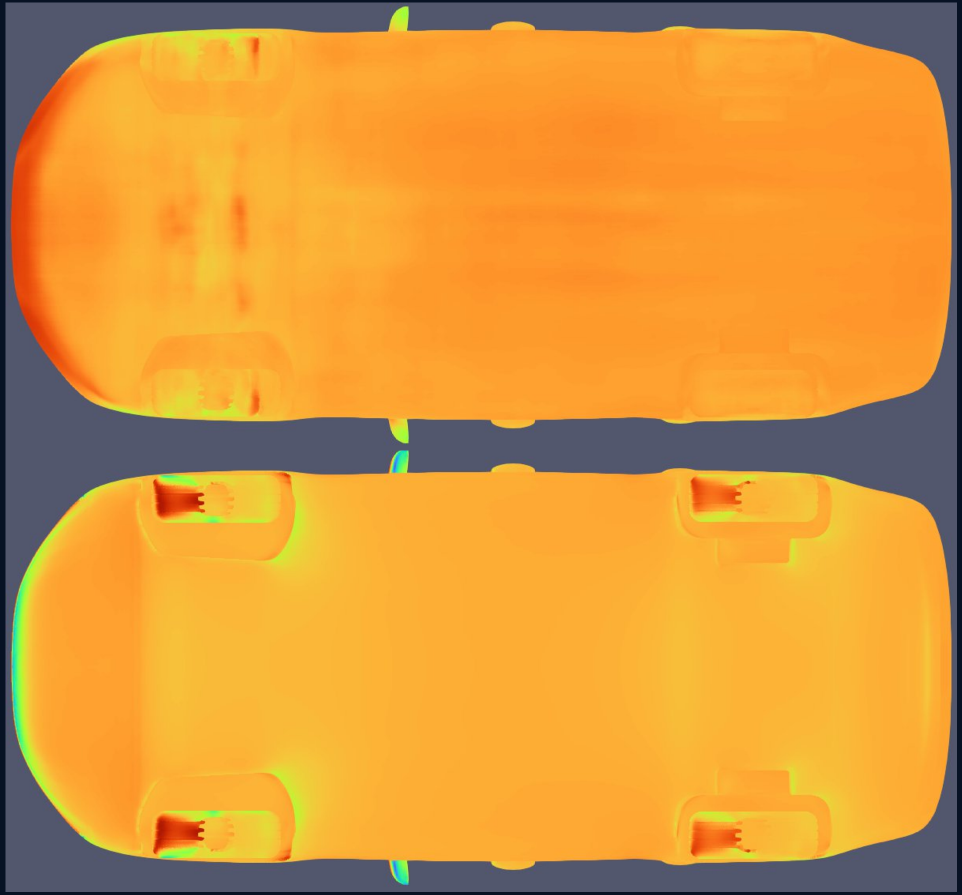
Work package 1.3: Evidence of the issue





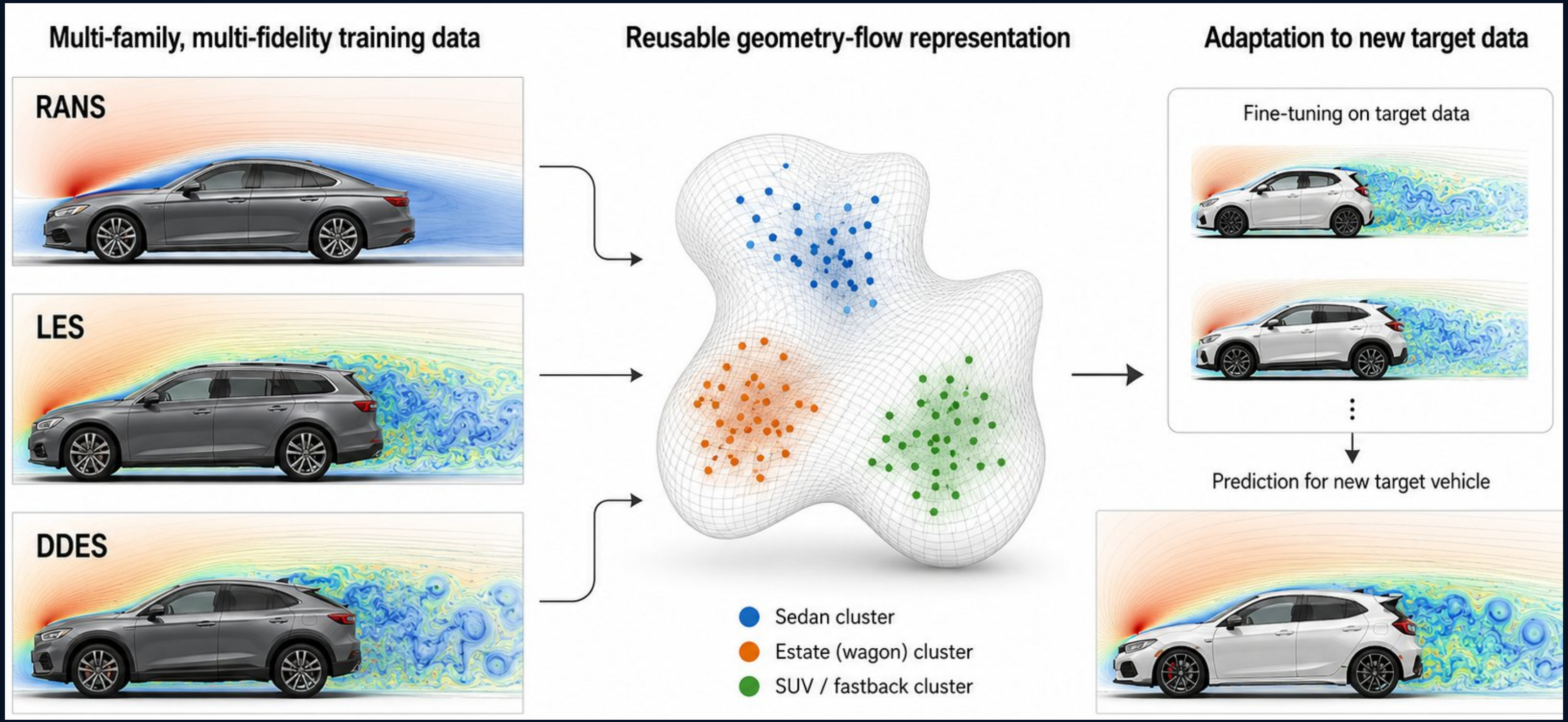
Prediction

CFD



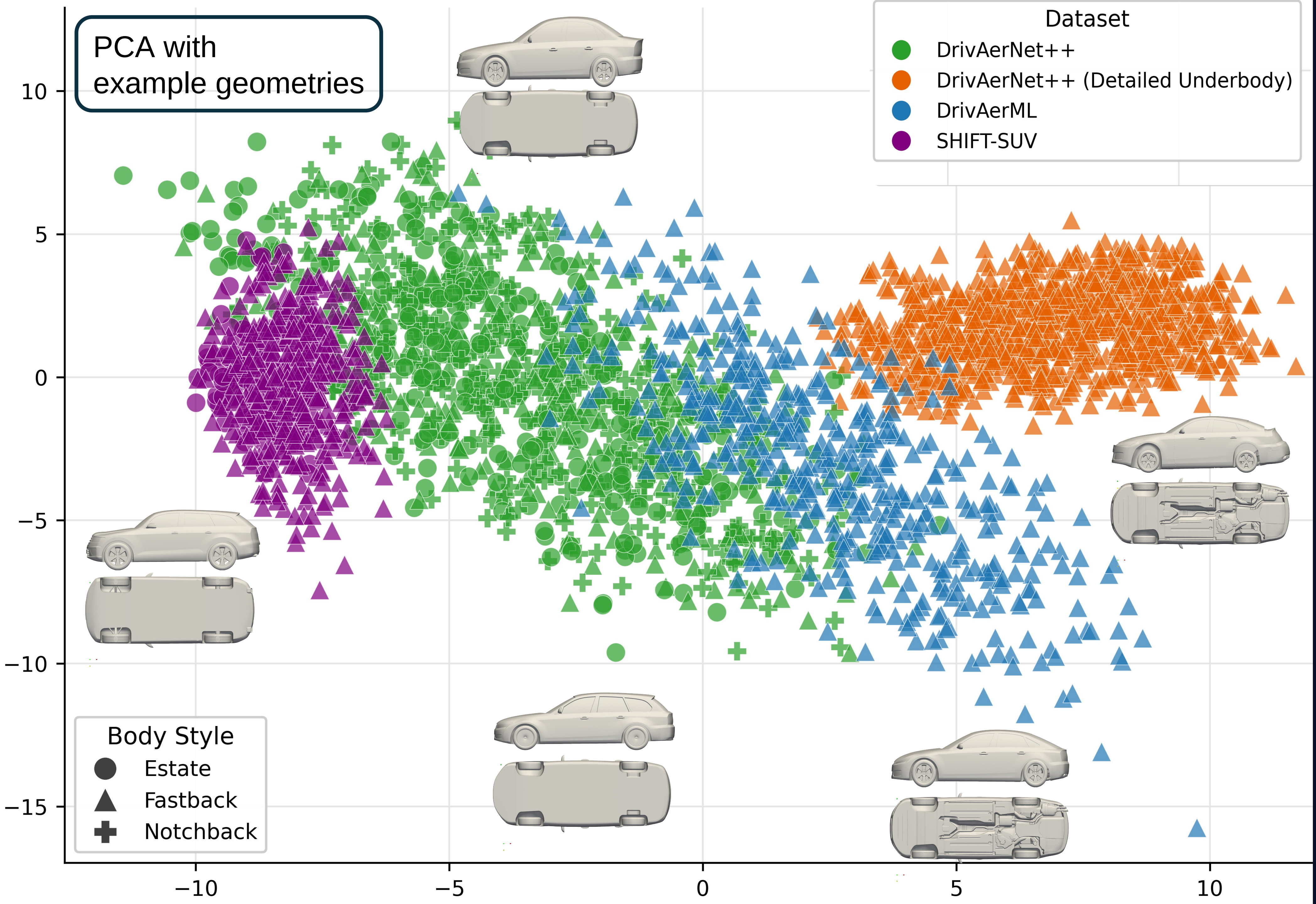
Objective 2 (long term): Foundation model and multi-fidelity learning

The later-term goal and direction is a reusable aerodynamic prediction model that can adapt itself to new data and solver methods.
It should be able to learn information across vehicle families and CFD fidelity levels .



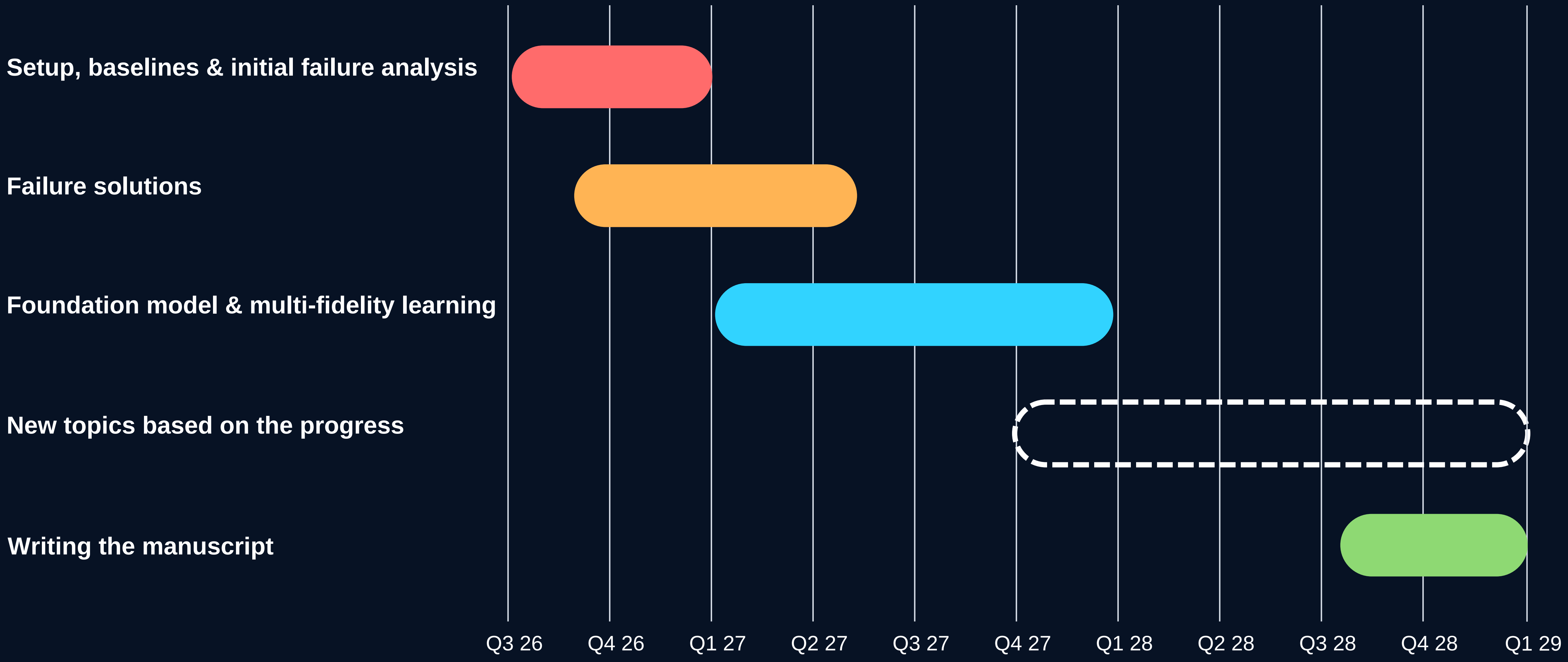
KEY PRINCIPLE

RANS, LES, and DDES should not be mixed as identical labels. The model should learn how fidelity levels relate to each other.


PCA with
example geometries
Dataset
DrivAerNet++
DrivAerNet++ (Detailed Underbody)
DrivAerML
SHIFT-SUV
Body Style
Estate
Fastback
Notchback
10
5
0
−5
−10
−15
−10
−5
0
5
10

TIMELINE

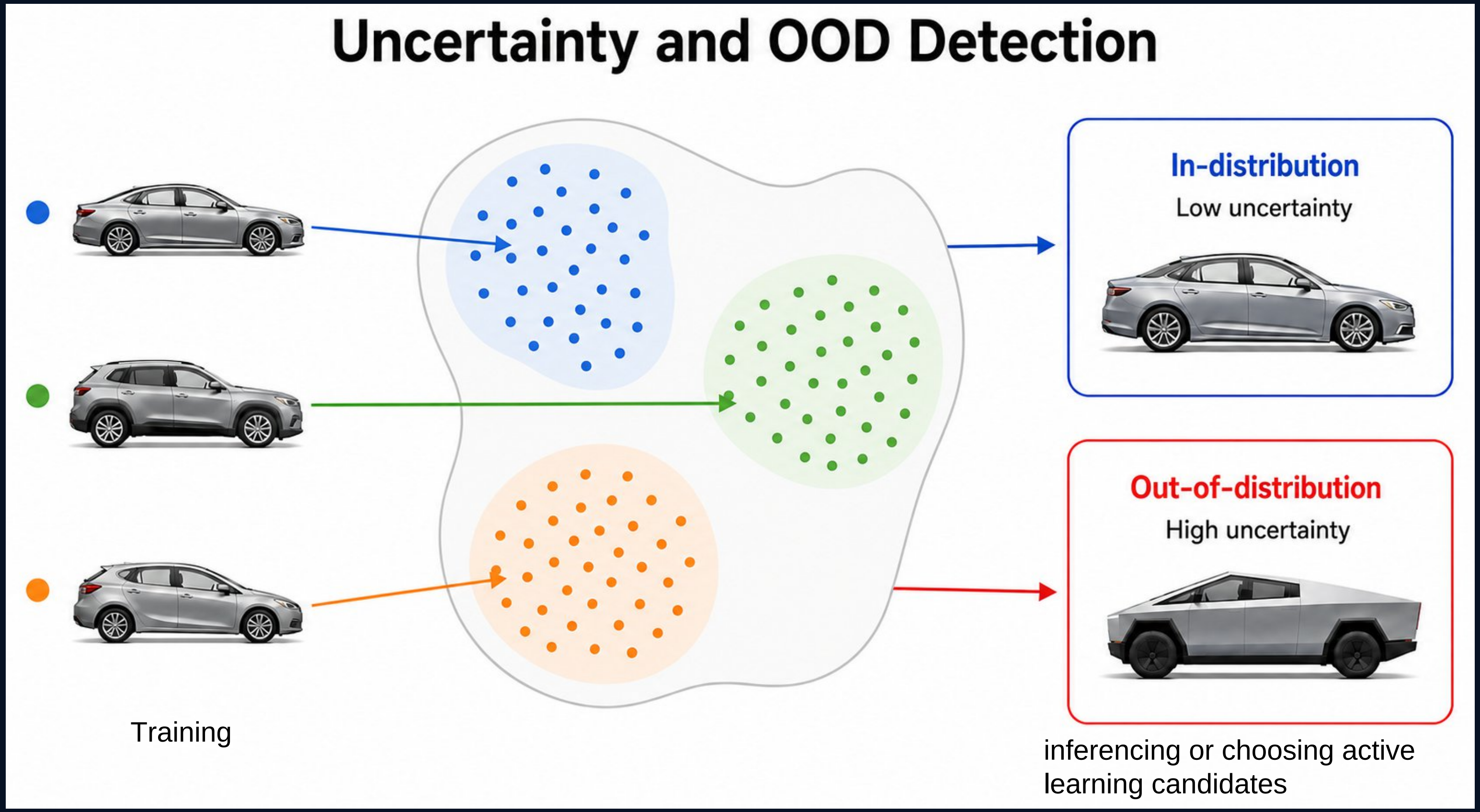
May 2026 - Jan 2029



Backup

Objective 3: Uncertainty and OOD Detection

For engineering use, the model must say more than what it predicts. It must also say how much the prediction should be trusted, especially for new geometries. This can also help generating fine-tuning candidates that truly fills the knowledge gap of the model.



Objective 4: Optimization

The optimization goal is to support aerodynamicists in exploring geometry changes faster. The model should understand how geometry changes affect aerodynamics and guide a geometry generative model accordingly